

Md Omar Faruqe Riyad

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EDUCATION

Shahjalal University of Science and Technology (SUST)

Sylhet, Bangladesh

B.Sc. in Computer Science and Engineering; CGPA: 3.53/4.00 (Last two years: 3.83)

2018 – 2023

- GRE: 317/340 (Q: 160, V: 157, AW: 3.0)
- TOEFL: 104/120 (R: 29, L: 27, S: 23, W: 25)

RESEARCH & PUBLICATIONS

AI for Site Reliability Engineering (SREGym) | *Research Intern, advised by Prof. Tianyin Xu* UIUC, 2026

- Contributing to **SREGym**, an open-source benchmark evaluating AI agents on realistic cloud-native system failures; conducting case studies and postmortem analysis of real-world outages projected into reproducible benchmark problems.
- Studied agent failure modes documented in the SREGym and Stratus papers (cross-layer reasoning gaps, greedy diagnosis anchoring, reward hacking).
- Designing fault scenarios and evaluation oracles that target these weaknesses and quantify where state-of-the-art SRE agents fail to reason across system layers, from control-plane admission to network policy.

AdversaryAI at BLP-2025 Task 2

BLP 2025, IJCNLP-AACL

- **Low-Resource Adaptation:** Investigated low-resource language code generation by fine-tuning open-source models (Llama 3, TigerLLM, Qwen) using parameter-efficient **LoRA** adapters.
- **Reasoning Enhancement:** Enhanced model reasoning via **Chain-of-Thought (CoT)** and a self-refinement loop that iteratively critiques and corrects generated code based on execution feedback.
- **Error Analysis:** Conducted rigorous error analysis to identify failure modes, achieving **85% accuracy** and **4th Place** out of 32 teams.

Deja Vu at SemEval 2024 Task 9

SemEval 2024, NAACL

- Designed data augmentation pipelines to improve model robustness for common sense reasoning tasks.
- Conducted a comparative study of Advanced Language Models, analyzing performance gaps in lateral thinking puzzles.

Team Syrax at BLP-2023 Task 1

BLP 2023, EMNLP

- **Semi-Supervised Learning:** Applied self-training techniques and back-translation to mitigate data scarcity and class imbalance in Bangla text.
- **Ensemble Strategy:** Implemented bagging and majority voting ensembles for Violence Inciting Text Detection, achieving **Top 20** rank.

Visual Question Answering (VQA) Data Synthesis | *Undergraduate Thesis*

2022

- Addressed "Answer Space Diversity" limitation in VQA v2 by augmenting training data with automatically generated, contextually relevant QA pairs.
- Conducted a comprehensive study of VQA methods to analyze limitations in visual reasoning.

EXPERIENCE

Dynamic Solution Innovators

April 2023 – Present

Software Engineer

Dhaka, Bangladesh

- Built and open-sourced **gf** , a GlassFish development CLI that **hot-swaps bytecode** into running JVMs via JDI/JDWP and bypasses JasperReports classloader caching, reducing local edit-deploy cycles from **~2 minutes to ~5 seconds**.
- Re-architected file storage and delivery behind a pluggable **StorageClient** abstraction supporting MinIO and filesystem backends, replacing base64-inlined file rendering with URL-based streaming to eliminate large HTML payloads and reduce server-side memory pressure. Added scoped file authorization and **nginx X-Accel-Redirect** offloading for efficient file serving.
- Designed a reusable **approval-workflow engine** backed by configurable state transitions, replacing hardcoded per-module approval logic. Engineered a **dynamic field-level correction system** enabling reviewers to reopen only selected form fields with inline feedback while preserving audit history.

- Designed a **JMS**-based notification pipeline with DLQ-backed retries, isolating SMS/email failures from the synchronous request path.
- Performed application security reviews and **static/dynamic testing against OWASP Top 10** risks as part of the internal Cybersecurity Team, catching access-control, injection, and validation flaws before production release.

Altri.ai

Full-Stack AI Engineer, RevestAI 

Part-time
Remote, US

- Rebuilt the LLM enrichment pipeline around `asyncio.gather` with bounded concurrency and atomic batch commits, allowing failed batches to resume without reprocessing completed LLM calls and improving throughput by **~10×** over the prior sequential pipeline.
- Designed the **three-stage property scoring pipeline** separating LLM feature extraction, valuation models, and deterministic financial heuristics, enabling each layer to be independently retrained and tested.
- Built the property ingestion pipeline on HomeHarvest and **PostgreSQL** using content-hash deduplication and incremental delta-syncing, preventing duplicate enrichment runs while preserving historical listing state across repeated scrapes.
- Implemented **distributed job coordination on PostgreSQL** using `SELECT FOR UPDATE` row-level locks with automatic stale-lock recovery, allowing multiple scheduler replicas to safely process long-running enrichment and training jobs.
- Built the ML training and promotion pipeline for **XGBoost**-based property valuation using Champion/Challenger validation and curated holdout datasets to prevent unverified models from triggering production-wide rescoring.

PROJECTS

SREGym Fault Scenario Contribution | Python, Kubernetes, Docker 2026

- Reproduced a documented Kubernetes admission-control incident ([issue #128162](#)) where multiple mutating admission webhooks' cumulative timeouts exceed the API server's global admission deadline, silently blocking all pod creation.
- Designed a four-property mitigation oracle with a fresh probe-pod admission test that rejects reward-hack fixes masking the symptom (e.g., deleting all webhooks) without restoring the required control-plane network path.

KnowledgeRelay | Python, FastAPI, LangChain, ChromaDB, React 2025

- Built a RAG-based knowledge transfer system during a 24-hour onsite hackathon, achieving **6th place** in the DSI AI Agent Hackathon 2025.
- Engineered a multi-stage ingestion pipeline that generates structured QA pairs from documents and supports context-aware conversational retrieval.

PrimeFaces Open Source Contribution | Java, JSF, PrimeFaces 2024

- Contributed a feature enhancement to PrimeFaces' `JPALazyDataModel`, enabling advanced custom filtering for DataTable-backed JPA queries.

HONORS, AWARDS & CERTIFICATIONS

4th Place out of 32 teams – Code Generation in Bangla Shared Task, IJCNLP-AAACL 2025 [\[Leaderboard\]](#)

6th Place out of 30 teams – DSI AI Agent Hackathon, 2025 [\[Leaderboard\]](#)

Top 20 (Rank 12 post-eval) – Violence Inciting Text Detection (VITD), BLP 2023 [\[Leaderboard\]](#)

Ranked 11th – BUET CSE Fest 2022 AI Contest

Certified in Cybersecurity (CC) – ISC2 [\[Badge\]](#)

Solved 500+ algorithmic problems on Codeforces, LeetCode, and LightOJ.

LEADERSHIP & TEACHING

Course Instructor 2024 – Present

Dynamic Learning

Remote

- Authored and instructed [Enterprise Web Development with JEE, JSF, and PrimeFaces](#) course, teaching core architectural patterns including component-driven design, CDI/EJB integration, and JSF lifecycle management. 

Volunteer 2019

- Served as volunteer at IEEE International Conference on Bangla Speech and Language Processing (ICBSLP).

TECHNICAL SKILLS

Languages: Java, Python, SQL, Bash, JavaScript, C/C++

Backend & Systems: Jakarta EE (JSF, CDI, EJB, JPA, JMS), Spring Boot, FastAPI, GlassFish, AspectJ, Kafka

AI & ML Systems: PyTorch, LLMs, RAG, LangChain, ChromaDB, XGBoost, LoRA Fine-tuning, NLP

Databases & Storage: PostgreSQL, MySQL, Redis, Neo4j, Supabase, MinIO, Hibernate/JPA, Flyway, Alembic

Security: OWASP Top 10, RBAC, Jakarta Security, JWT

Infrastructure & Observability: Docker, Kubernetes, Git, GitHub Actions, Prometheus, Grafana, Loki, Nginx, Ansible, ZFS

Frontend: React, PrimeFaces